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A parareal exponential integrator finite element method for semilinear parabolic equations

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Abstract

In this article, we present a parareal exponential finite element method, with the help of variational formulation and parareal framework, for solving semilinear parabolic equations in rectangular domains. The model equation is first discretized in space using the finite element method with continuous piecewise multilinear rectangular basis functions, producing the semi-discrete system. We then discretize the temporal direction using the explicit exponential Runge–Kutta approach accompanied by the parareal framework, resulting in the fully-discrete numerical scheme. To further improve computational speed, we design a fast solver for our method based on tensor product spectral decomposition and fast Fourier transform. Under certain regularity assumption, we successfully derive optimal error estimates for the proposed parallel-based method with respect to H^1 -norm. Extensive numerical experiments in two and three dimensions are also carried out to validate the theoretical results and demonstrate the performance of our method.

KEYWORDS

error estimates, exponential integrator, finite element method, parareal method, Runge–Kutta, semilinear parabolic equations

1 | INTRODUCTION

Semilinear parabolic equations are extensively used in mathematical models arising from scientific and engineering applications, such as the Allen–Cahn [1] equations for modeling phase transitions

and separations, and the time-dependent advection diffusion and Navier–Stokes equations [22] for fluid dynamics, among others. To satisfy various realistic demands, the development of efficient and accurate numerical schemes for semilinear parabolic equations has been widely studied in recent years. In this article, we focus primarily on a family of second-order semilinear parabolic equations given by

$$u_t = D\Delta u + f(t, u), \quad x \in \Omega, \quad t > 0, \quad (1)$$

where Ω is an open rectangular domain in $\mathbb{R}^d$ ($d \geq 1$), $D > 0$ is the diffusion coefficient, $u(t, x) : \Omega \rightarrow \mathbb{R}$ is the unknown function, and $f(t, u)$ is the reaction term.

There are numerous numerical methods available for solving semilinear parabolic equations of the form (1). Temporal discretization schemes include the implicit-explicit method [2], split-step method [6, 35], integrating factor method [31], sliders method [11], exponential time differencing method (ETD) [5, 26], scalar auxiliary variable method (SAV) [29, 36], integrating factor Runge–Kutta (IFRK) method [28, 32], and others. Spatial discretization is typically implemented using finite element method, finite difference method, finite volume method, or spectral method. The convergence analysis highly depends on discrete maximum bound principles, which does not always hold for the classic finite element or pseudo-spectral discretizations, see [12, 30] and references cited therein. In our recent work [27], we introduced the exponential integrator finite element method (EIFE) which combines the finite element method in space and explicit exponential Runge–Kutta method in time for solving semilinear parabolic equations. Rigorous error estimates were successfully derived and a FFT-based fast solver was also provided based on tensor product spectral decomposition when the Dirichlet or periodic boundary conditions are imposed.

Modern computer architecture emphasizes the concurrency of processor groups, rather than the computing capacity improvement of single processor. In this regard, it is essential and highly desired to develop efficient parallel algorithms to make use of high concurrency of processor groups for large-scale systems. The parareal algorithm has been such a crucial method since it was proposed by Lions, Maday, and Turinici to solve ordinary differential equations (ODEs) in the parallel-in-time pattern [34]. The parareal algorithm is composed of a coarse propagator and a fine propagator and makes use of the iterative approach, and upon convergence, the approximation on the coarse time grid will achieve the same accuracy as that of using the fine propagator step by step with the small time step size. The convergence property of the parareal algorithm is closely related to the coarse and fine propagators. It was proved in [19] that the parareal algorithm with an exact fine propagator is superlinearly convergent on bounded time intervals and linearly convergent for unbounded intervals. By constructing some Banach spaces, a more accurate estimation of stability and convergence was obtained in [3]. Stability of the parareal algorithm for the equations with complex eigenvalues was discussed in [37, 43] with the help of the stability function. A posteriori error estimation was also studied for scalar ODEs in [8]. Parareal-Richardson and Parareal-IMEX RK methods have been developed to achieve higher-order accuracy and better numerical stability [39, 42], and the convergence rate of several second-order parareal methods was analyzed in [40]. In recent years, significant efforts have been devoted to developing modified parareal algorithms, which have demonstrated better convergence rates and substantial speedup [13–15, 19–21, 23]. As is well-known, the coarse grid correction (CGC) typically limits the parallel efficiency of parareal methods. To overcome this limitation, researchers have explored various modifications to the parareal approach, including reducing iteration times [21], decoupling subproblems [16], diagonalizing ODEs [18, 41], and more. Additionally, the parareal algorithm has been extended to handle non-differentiable PDEs [4] and scalar time-periodic ODEs [17].

In this article, we propose an efficient method for solving the semilinear parabolic equation (1) in rectangular domains, called the parareal exponential integrator finite element method (Parareal-EIFE), which combines the finite element method for spatial discretization and explicit exponential Runge–Kutta approach in the parareal framework for temporal discretization. It is essentially a modified parareal method with the EIFE coarse and fine propagators. Building upon the ideas presented in [25] and [3, 19], we show that the proposed Parareal-EIFE method can achieve the accuracy of applying the fine propagator with fine step size sequentially. As a result, explicit error estimates in the H^1 -norm are derived for the Parareal-EIFE method with one and two inner stages for two-level propagators with Dirichlet boundary conditions. Furthermore, the Parareal-EIFE method also can be efficiently implemented using FFT, as the resulting mass and coefficient matrices of the semi-discrete system can be diagonalized simultaneously. To the best of our knowledge, this article presents the first study on a fast numerical method in parallel with rigorous error estimates, which combines exponential integrator and parareal framework in time, and finite element discretization in space.

The rest of the article is structured as follows. In Section 2, we present the Parareal-EIFE method, an efficient and accurate approach for solving semilinear parabolic equations with the Dirichlet or periodic boundary conditions. In Section 3, we review some assumptions, lemmas, and the error bounds of applying the EIFE method to a single time subinterval. In Section 4, convergence analysis is performed to derive optimal error estimates for the proposed Parareal-EIFE method. In Section 5, we numerically demonstrate the performance of the Parareal-EIFE method through various experiments. Finally, some concluding remarks are drawn in Section 6.

2 | THE PARAREAL EXPONENTIAL INTEGRATOR FINITE ELEMENT METHOD

In this section, we will first review the exponential integrator finite element method proposed in [27] for the semilinear parabolic equation (1). Then, we will develop a parareal algorithm based this method to carry out parallel implementation in time.

First of all, some standard notations are introduced for later requirement. For a given bounded Lipschitz domain $\Omega \subset \mathbb{R}^d$ and nonnegative integer s , denote $H^s(\Omega)$ as the standard Sobolev spaces on domain Ω with norm $\|\cdot\|_{s,\Omega}$ and semi-norm $|\cdot|_{s,\Omega}$, and the corresponding L^2 -inner product is $(\cdot, \cdot)_{\Omega}$. Also, we denote $H_0^s(G)$ as the closure of $C_0^\infty(G)$ with respect to the norm $\|\cdot\|_{s,G}$ and $\|v\|_{k,\infty,G} = \text{ess sup}_{|\alpha| \leq k} \|D^\alpha v\|_{L^\infty(G)}$ for any function v such that the right-hand side term makes sense, where $\alpha = (\alpha_1, \dots, \alpha_d)$ is a multi-index and $|\alpha| = \alpha_1 + \dots + \alpha_d$. Generally speaking, we omit the subscript for simplicity if there is no confusion. For a non-negative integer ℓ , the set of all polynomials on Ω with the total degree at most ℓ are denoted as $P_\ell(\Omega)$. Moreover, given two quantities a and b , “ $a \lesssim b$ ” is the abbreviation of “ $a \leq Cb$,” where the hidden constant C is positive and independent of the mesh size; $a \approx b$ is equivalent to $a \lesssim b \lesssim a$.

Let us consider the model equation (1) equipped with a homogeneous Dirichlet boundary condition and an initial condition $u_0 \in H^2(\Omega) \cap H_0^1(\Omega)$, that is

$$\begin{cases} \partial_t u = D\Delta u + f(t, u), & \mathbf{x} \in \Omega, \ 0 < t \leq T, \\ u(0, \mathbf{x}) = u_0(\mathbf{x}), & \mathbf{x} \in \Omega, \\ u(t, \mathbf{x}) = 0, & \mathbf{x} \in \partial\Omega, \ 0 < t \leq T, \end{cases} \quad (2)$$

where $T \geq 0$ is the terminal time.

2.1 | Semi-discretization in space by finite element approximation

Assume that the rectangular domain $\Omega \in \mathbb{R}^d$ can be represented as $\bar{\Omega} = \prod_{i=1}^d [a_i, b_i]$. The domain Ω is divided uniformly along direction x_i with mesh size $h_i = \frac{b_i - a_i}{N_i + 1}$ for $1 \leq i \leq d$ with $N_i > 0$. Then, the finite element space V_h can be constructed by using the tensor product of all finite element spaces $V_{h_i}^i(a_i, b_i)$ for direction x_i with one-dimensional continuous piecewise linear basis functions. Therefore, the finite element space V_h can be represented as

$$V_h := V_{h_1}^1(a_1, b_1) \otimes \cdots \otimes V_{h_d}^d(a_d, b_d) = \text{span} \left\{ \phi_1^{i_1}(x_1) \cdots \phi_d^{i_d}(x_d) : 1 \leq i_1 \leq N_1, \dots, 1 \leq i_d \leq N_d \right\}, \quad (3)$$

where $\phi_j^i(x_i)$ ($1 \leq j \leq N_i$) is the j th nodal basis function of $V_{h_i}^i(a_i, b_i)$. Let $h = \max_{1 \leq i \leq d} h_i$ denote the mesh size of the corresponding uniform rectangular partition $\mathcal{T}_h$ for generating V_h . For convenience, we always assume the partition $\mathcal{T}_h$ is quasi-uniform, that is, $h \approx h_i$ for all $1 \leq i \leq d$.

Based on the definition of V_h , after variational formulation and finite dimensional approximation, the original problem (2) can be reformulated as to find $u_h \in L^2(0, T; V_h)$ satisfying

$$\begin{cases} ((u_h)_t, v_h) + a(u_h, v_h) = (f(t, u_h), v_h), & \forall v_h \in V_h, 0 \leq t \leq T, \\ u_h(0) = P_h u_0, \end{cases} \quad (4)$$

where the bilinear operator $a(\cdot, \cdot)$ is defined by

$$a(w, v) = \int_{\Omega} D \nabla w \cdot \nabla v dx, \quad \forall w, v \in H_0^1(\Omega), \quad (5)$$

and $P_h : L^2(\Omega) \rightarrow V_h$ is the L^2 -orthogonal projection operator and is stable with respect to L^2 -norm and H^1 -norm [7], that is, $\|P_h u\|_0 \lesssim \|u\|_0$ and $\|P_h u\|_1 \lesssim \|u\|_1$ for any $u \in H_0^1(\Omega)$. According to the inverse inequality for finite elements and Riesz representation theorem, there exists a unique and bounded linear operator $L_h : V_h \rightarrow V_h$ such that

$$a(w_h, v_h) = (L_h w_h, v_h), \quad \forall w_h, v_h \in V_h. \quad (6)$$

Note that $(-L_h)$ is the infinitesimal generator of the semigroup $\{e^{-tL_h}\}_{t \geq 0}$ on V_h . Making use of the projector P_h and the operator L_h defined above, we can reformulate the problem (4) as the following equivalent semi-discrete (in space) system:

$$\begin{cases} (u_h)_t + L_h u_h = P_h f(t, u_h), & x \in \Omega, 0 < t \leq T, \\ u_h(0) = P_h u_0, & x \in \Omega. \end{cases} \quad (7)$$

2.2 | Explicit exponential Runge–Kutta integrator in time

For brevity, we always denote $u(t)$ as $u(t, \cdot)$ and $u_h(t)$ as $u_h(t, \cdot)$ for later analysis. The time interval $[0, T]$ is divided into $N_T > 0$ subintervals, $[\tau_n, \tau_{n+1}]$ for $n = 0, 1, \dots, N_T - 1$, where the nodes are formed by $\tau_{n+1} = \tau_n + \Delta \tau_n$ such that $\sum_{n=0}^{N_T-1} \Delta \tau_n = T$. By the Duhamel's formula, the semi-discrete system (7) is equivalent to

$$u_h(\tau_{n+1}) = e^{-\Delta \tau_n L_h} u_h(\tau_n) + \int_0^{\Delta \tau_n} e^{-(\tau_n - \sigma) L_h} P_h f(\tau_n + \sigma, u_h(\tau_n + \sigma)) d\sigma. \quad (8)$$

Applying the consistent explicit exponential Runge–Kutta method [26] to (8), we then obtain the EIFE method developed in [27], that is: given $u_h^0 = u_h(0)$, for $n = 0, 1, \dots, N_T - 1$, compute

$$u_h^{n+1} = e^{-\Delta\tau_n L_h} u_h^n + \tau_n \sum_{i=1}^s b_i(-\Delta\tau_n L_h) G_{ni}(u_h^n, \tau_n), \quad (9a)$$

$$U_{n,i}(u_h^n, \tau_n) = e^{-c_i \Delta\tau_n L_h} u_h^n + \tau_n \sum_{j=1}^{i-1} a_{ij}(-\Delta\tau_n L_h) G_{nj}(u_h^n, \tau_n), \quad i = 1, \dots, s, \quad (9b)$$

$$G_{nj}(u_h^n, \tau_n) = P_h f(\tau_n + c_j \tau_n, U_{n,j}(u_h^n, \tau_n)), \quad i = 1, \dots, s, \quad (9c)$$

where u_h^n is the numerical solution at time τ_n , $s > 0$ denotes the number of stages, the interpolation nodes $\{c_1, \dots, c_s\}$ are different points selected in $[0, 1]$ in the nondecreasing order, and the weights are defined by

$$\begin{cases} b_i(-\Delta\tau_n L_h) = \int_0^1 e^{-\Delta\tau_n(1-\theta)L_h} l_i(\theta) d\theta, & i = 1, \dots, s \\ a_{ij}(-\Delta\tau_n L_h) = \frac{1}{\tau_n} \int_0^{c_i \Delta\tau_n} e^{-(c_i \Delta\tau_n - \sigma)L_h} l_j(\sigma) d\sigma, & i = 1, \dots, s, j = 1, \dots, i-1, \end{cases} \quad (10)$$

with $\{l_i(\theta)\}_{i=1}^s$ denoting the classic Lagrange interpolation polynomials

$$l_i(\theta) := \prod_{m=1, m \neq i}^s \frac{\theta - c_m}{c_i - c_m}, \quad i = 1, \dots, s.$$

In particular, EIFE1 ($s = 1$) method is the first-order EIFE method with $c_1 = 0$ while EIFE2 ($s = 2$) is second-order with interpolation nodes $c_1 = 0$ and $c_2 \in (0, 1]$. We omit the details due to the space limitation and the details can be found in [27]. It's worth noting that the semi-discrete system (7) can be regarded as an ODE system, which motivates us to implement the EIFE method in parallel-in-time pattern.

2.3 | The EIFE-based parareal algorithm

Based on the EIFE method for the semilinear parabolic equation (1), we intend to develop its parareal version, Parareal-EIFE, that is, the parallel in time version of the fully-discrete scheme (9).

To describe the proposed parareal algorithm, we need some additional notations. Suppose that the whole time interval $[0, T]$ is divided into N coarse time subintervals and each coarse subinterval is denoted as $[T_n, T_{n+1}]$ with the time step size $\Delta T_n = T_{n+1} - T_n$ for $n = 0, 1, \dots, N - 1$. Next each of the coarse time intervals is further divided into M fine time intervals, which gives us a fine partition of the whole time domain $[0, T]$ as $[t_l, t_{l+1}]$ for $l = 0, 1, \dots, NM - 1$ with the fine time step size $\Delta t_l = t_{l+1} - t_l$. Clearly, $T_n = t_{nM}$ for $n = 0, 1, \dots, N - 1$. The coarse grids and fine grids are attached with a less accurate but cheaper propagator $\mathcal{G}$ and an accurate but expensive propagator $\mathcal{F}$, respectively. More specifically, we choose the EIFE method to construct both the coarse propagator $\mathcal{G}$ and the fine propagator $\mathcal{F}$. Taking v as the starting value, denote $\mathcal{G}(v; T_n, T_{n+1})$ as the result computed in the coarse time interval $[T_n, T_{n+1}]$ by the coarse propagator $\mathcal{G}$ with p stages and $\mathcal{F}(v; t_l, t_{l+1})$ as the result computed in the fine time interval $[t_l, t_{l+1}]$ calculated by the fine propagator $\mathcal{F}$ with q stages, that is,

$$\begin{cases} \mathcal{G}(v; T_n, T_{n+1}) = e^{-\Delta T_n L_h} v + \Delta T_n \sum_{i=1}^p b_i(-\Delta T_n L_h) G_{ni}(v, \Delta T_n), \\ \mathcal{F}(v; t_l, t_{l+1}) = e^{-\Delta t_l L_h} v + \Delta t_l \sum_{i=1}^q b_i(-\Delta t_l L_h) G_{ni}(v, \Delta t_l), \end{cases}$$

Algorithm 1. Parareal-EIFE method

- 1: Set $k = 0$ and $U_h^{0,(0)} = U_h^0 = P_h u_0$.
- 2: $\ominus$ For $n = 0, 1, \dots, N-1$, compute (by the coarse propagator) sequentially

$$U_h^{n+1,(0)} = \mathcal{G}(U_h^{n,(0)}; T_n, T_{n+1}).$$
- 3: **while** the stopping criteria is not satisfied **do**
- 4: $\oplus$ For $n = 0, 1, \dots, N-1$, compute (by the fine propagator) in parallel

$$\bar{U}_h^{n+1,(k+1)} = \mathcal{F}^M(U_h^{n,(k)}; T_n, T_{n+1}).$$
- 5: $\ominus$ For $n = 0, 1, \dots, N-1$, compute sequentially

$$U_h^{n+1,(k+1)} = \mathcal{G}(U_h^{n,(k+1)}; T_n, T_{n+1}) + \bar{U}_h^{n+1,(k+1)} - \mathcal{G}(U_h^{n,(k)}; T_n, T_{n+1}), \quad (11)$$
 with $U_h^{0,(k+1)} = U_h^0$.
- 6: Set $k = k + 1$.
- 7: **end while**
- 8: Set $U_h^n = U_h^{n,(k)}$ for $n = 1, 2, \dots, N$ and return $\{U_h^n\}_{n=0}^N$.

with G_{ni} defined in (9). We also denote $\mathcal{F}^M(v; T_n, T_{n+1})$ as the result produced by applying successively M substeps of the fine propagator $\mathcal{F}$ in the coarse time interval $[T_n, T_{n+1}]$. To analyze the error estimate of Parareal-EIFE method more clearly, we denote $\mathcal{H}(v; T_n, T_{n+1})$ as the exact solution produced in the time interval $[T_n, T_{n+1}]$ with the initial value v . Denote $U_h^{n,(k)}$ as the numerical result computed for T_n after k iterations and U_h^n as the final approximate solution for T_n . To better illustrate the Parareal-EIFE method, we designate the time-sequential parts of the algorithm with symbol $\ominus$, and the time-parallel parts with symbol $\oplus$. The framework of our Parareal-EIFE method is shown in Algorithm 1.

As illustrated in [3], the convergence order of a parareal method depends mainly on that of the fine propagator once it is convergent. Consequently, to balance the computational cost and accuracy of Algorithm 1, one better chooses a more accuracy EIFE method with more inner stages for the fine propagator $\mathcal{F}$ and a less accurate one with less inner stages for the coarse propagator $\mathcal{G}$, that is, $p \leq q$. It is worthy noting that the Parareal-EIFE method can be realized in fast implementation by FFT by following the procedure given in [27]. In addition, the Parareal-EIFE method can be applied to the model problem with the periodic boundary condition as shown in [27].

3 | PRELIMINARY RESULTS

In this section, we will recall some assumptions and lemmas which have been presented in [27]. Also, we will discuss the error estimates of applying the EIFE method to a single time subinterval. Moreover, we also assume $d \leq 3$ from now on. We will perform the error analysis of the Parareal-EIFE method by following the similar arguments in [25] and [3, 20].

3.1 | Useful lemmas

We first recall some fundamental estimates for the semigroup $\{e^{-tL_h}\}_{t \geq 0}$.

Lemma 1 ([27]). *Let L_h be the linear operator defined by (6).*

(i) *For any given parameter $\gamma \geq 0$, it holds*

$$\|e^{-tL_h}\|_0 + \|t^\gamma L_h^\gamma e^{-tL_h}\|_0 \lesssim 1, \quad \forall t \geq 0, \forall h > 0. \quad (12)$$

(ii) For any given parameter $0 \leq \gamma \leq 1$, it holds

$$\|\phi(-tL_h)\|_0 + \|t^\gamma L_h^\gamma \phi(-tL_h)\|_0 \lesssim 1, \quad \forall t > 0, \forall h > 0, \quad (13)$$

where $\phi(-tL_h) = b_i(-tL_h)$ or $\phi(-tL_h) = a_{ij}(-tL_h)$, $i, j = 1, \dots, s$.

(iii) For any given parameter $0 \leq \gamma \leq 1$, it holds

$$\left\| t L_h^\gamma \sum_{j=1}^{n-1} e^{-j t L_h} \right\|_0 \lesssim 1, \quad \forall t > 0, \forall h > 0. \quad (14)$$

By the definitions (5) and (6) and noting that L_h is symmetric and positive definite, we have the following important property for L_h :

$$\|v\|_1 \approx \left\| L_h^{\frac{1}{2}} v \right\|_0, \quad \forall v \in V_h, \quad (15)$$

which establishes an important relation between H^1 and L^2 norms on the finite element space V_h . Also, as given in [38], we introduce the mild growth condition for the function f and some regularity conditions required for the exact solution $u(t)$ of (2) in order to carry out convergence and error analysis of the Parareal-EIFE method.

Assumption 3.1 ([38]). The function $f(t, \zeta)$ grows mildly with respect to ζ , that is, there exists a number $p > 0$ for $d = 1, 2$ or $p \in (0, 2]$ for $d = 3$ such that

$$\left| \frac{\partial f}{\partial \zeta}(t, \zeta) \right| \lesssim 1 + |\zeta|^p, \quad \forall t, \zeta \in \mathbb{R}. \quad (16)$$

Assumption 3.2 ([38]). The function $f(t, \zeta)$ is sufficiently smooth with respect to t and ζ , that is, for any given constant $K > 0$, it holds

$$\sum_{|\alpha| \leq 2} \left| D^\alpha f(t, \zeta) \right| \lesssim 1, \quad \forall t \in [0, T], \zeta \in [-K, K]. \quad (17)$$

Assumption 3.3 ([38]). The exact solution $u(t)$ satisfies some of the following regularity conditions

$$\sup_{0 \leq t \leq T} \|u(t)\|_{2, \Omega} \lesssim 1, \quad (18a)$$

$$\sup_{0 \leq t \leq T} \|u_t(t)\|_{0, \infty, \Omega} \lesssim 1, \quad (18b)$$

$$\sup_{0 \leq t \leq T} \|u_{tt}(t)\|_{0, \infty, \Omega} \lesssim 1, \quad (18c)$$

where the hidden constants may depend on T .

In addition, we also have the following important result on f .

Lemma 2 ([27]). Suppose that the function f satisfies Assumption 3.1, and the exact solution $u(t)$ fulfills (18a) in Assumption 3.3. Then f is locally-Lipschitz continuous in a strip along the exact solution $u(t)$, that is, for any given constant $R > 0$,

$$\|f(t, v(t)) - f(t, w(t))\|_0 \lesssim \|v(t) - w(t)\|_1, \quad (19)$$

for any $t \in [0, T]$ and $v(t), w(t) \in V_h$ satisfying

$$\max\{\|v(t) - u(t)\|_1, \|w(t) - u(t)\|_1\} \leq R,$$

where the hidden constant in (19) may depend on R .

The following result readily comes from Theorem 14.2 in [38].

Lemma 3. Suppose that the function f satisfies Assumptions 3.1 and 3.2, and the exact solution $u(t)$ fulfills (18a) in Assumption 3.3. There exists a constant $h_0 > 0$ such that if the spatial mesh size $h \leq h_0$, then

$$\|u_h(t) - u(t)\|_1 \lesssim h, \quad \forall t \in [0, T], \quad (20)$$

where the hidden constant is independent of h .

3.2 | Error bounds for a single time subinterval

For simplicity, we assume the time subintervals are uniform, that is, $\Delta t = \Delta t_0 = \dots = \Delta t_{NM-1} = \frac{T}{NM}$ and $\Delta T = \Delta T_0 = \dots = \Delta T_{N-1} = \frac{T}{N}$. Since the coarse propagator and the fine propagator are both constructed through the EIFE method (9), for brevity, we will denote the time interval as $[\tau_n, \tau_{n+1}]$ with the uniform time step size $\Delta\tau, n = 0, \dots, N_T - 1$ and present the error bounds for a single time subinterval. We can convert the solution (8) into the sum of the following two parts for further analysis:

$$\begin{aligned} u_h(\tau_{n+1}) &= e^{-\Delta\tau L_h} u_h(\tau_n) + \int_0^{\Delta\tau} e^{-(\Delta\tau-\sigma)L_h} P_h f(\tau_n + \sigma, u_h(\tau_n + \sigma)) d\sigma \\ &= e^{-\Delta\tau L_h} u_h(\tau_n) + \int_0^{\Delta\tau} e^{-(\Delta\tau-\sigma)L_h} P_h f(\tau_n + \sigma, u(\tau_n + \sigma)) d\sigma \\ &\quad + \int_0^{\Delta\tau} e^{-(\Delta\tau-\sigma)L_h} (P_h f(\tau_n + \sigma, u_h(\tau_n + \sigma)) - P_h f(\tau_n + \sigma, u(\tau_n + \sigma))) d\sigma. \end{aligned} \quad (21)$$

For brevity, let us define the following functions

$$\begin{cases} \varphi_i(-\Delta\tau L_h) = \frac{1}{i!} \int_0^{\Delta\tau} e^{-(\Delta\tau-\sigma)L_h} \frac{\sigma^{i-1}}{(i-1)!} d\sigma, i = 1, \dots, s, \\ \psi_i(-\Delta\tau L_h) = \varphi_i(-\Delta\tau L_h) - \sum_{k=1}^s b_k(-\Delta\tau L_h) \frac{c_k^{i-1}}{(i-1)!}, i = 1, \dots, s \\ \psi_{j,i}(-\Delta\tau L_h) = \varphi_j(-c_i \Delta\tau L_h) c_i^j - \sum_{k=1}^{i-1} a_{ik}(-\Delta\tau L_h) \frac{c_k^{j-1}}{(j-1)!}, i, j = 1, \dots, s. \end{cases} \quad (22)$$

Denote $f^{(k)}(t, u(t)) = \frac{d^k}{dt^k} f(t, u(t))$ as the k th full differentiation of f with respect to t . Comparing the numerical scheme (9) with (21), we can derive the following results according to Taylor expansion,

$$u_h(\tau_n + c_i \Delta\tau) = e^{-c_i \Delta\tau L_h} u_h(\tau_n) + \Delta\tau \sum_{j=1}^{i-1} a_{ij}(-\Delta\tau L_h) P_h f(\tau_n + c_j \Delta\tau, u(\tau_n + c_j \Delta\tau)) + \delta_{ni}, \quad (23)$$

$$u_h(\tau_{n+1}) = e^{-\Delta\tau L_h} u_h(\tau_n) + \Delta\tau \sum_{i=1}^s b_i(-\Delta\tau L_h) P_h f(\tau_n + c_i \Delta\tau, u(\tau_n + c_i \Delta\tau)) + \delta_{n+1}, \quad (24)$$

where the defects $\{\delta_{ni}\}_{i=1}^s, \delta_{n+1}$ are respectively given by

$$\begin{aligned} \delta_{ni} &= \sum_{j=1}^r (\Delta\tau)^j \psi_{j,i}(-\Delta\tau L_h) P_h f^{(j-1)}(\tau_n, u(\tau_n)) + \delta_{ni}^{[r]}, \\ \delta_{n+1} &= \sum_{j=1}^r (\Delta\tau)^j \psi_j(-\Delta\tau L_h) P_h f^{(j-1)}(\tau_n, u(\tau_n)) + \delta_{n+1}^{[r]} \end{aligned}$$

with the remainders $\delta_{ni}^{[r]}$ and $\delta_{n+1}^{[r]}$ defined respectively by

$$\begin{aligned}\delta_{ni}^{[r]} &= \int_0^{c_i \Delta \tau} e^{-(c_i \Delta \tau - \sigma) L_h} \int_0^\sigma \frac{(\sigma - \eta)^{r-1}}{(r-1)!} P_h f^{(r)}(\tau_n + \eta, u(\tau_n + \eta)) d\eta d\sigma \\ &\quad - \Delta \tau \sum_{k=1}^{i-1} a_{ik} (-\Delta \tau L_h) \int_0^{c_k \Delta \tau} \frac{(c_k \Delta \tau - \sigma)^{r-1}}{(r-1)!} P_h f^{(r)}(\tau_n + \sigma, u(\tau_n + \sigma)) d\sigma \\ &\quad + \int_0^{c_i \Delta \tau} e^{-(c_i \Delta \tau - \sigma) L_h} (P_h f(\tau_n + \sigma, u_h(\tau_n + \sigma)) - P_h f(\tau_n + \sigma, u(\tau_n + \sigma))) d\sigma \\ \delta_{n+1}^{[r]} &= \int_0^{\Delta \tau} e^{-(\Delta \tau - \sigma) L_h} \int_0^\sigma \frac{(\sigma - \eta)^{r-1}}{(r-1)!} P_h f^{(r)}(\tau_n + \eta, u(\tau_n + \eta)) d\eta d\sigma \\ &\quad - \Delta \tau \sum_{i=1}^s b_i (-\Delta \tau L_h) \int_0^{c_i \Delta \tau} \frac{(c_i \Delta \tau - \sigma)^{r-1}}{(r-1)!} P_h f^{(r)}(\tau_n + \sigma, u(\tau_n + \sigma)) d\sigma \\ &\quad + \int_0^{\Delta \tau} e^{-(\Delta \tau - \sigma) L_h} (P_h f(\tau_n + \sigma, u_h(\tau_n + \sigma)) - P_h f(\tau_n + \sigma, u(\tau_n + \sigma))) d\sigma.\end{aligned}$$

Here, r can be any nonnegative integer such that $f^{(r)}(t, u(t))$ exists and is continuous.

Assume that $\bar{v}_n$ is the result produced at τ_n by applying the EIFE scheme (9) for $(n-j)$ steps with the initial value $v_j \in V_h$ at τ_j and the time step size $\Delta \tau$. Define the numerical errors

$$\begin{aligned}E_{n,i}(v_j, \Delta \tau) &= U_{n,i}(v_j, \Delta \tau) - u_h(\tau_n + c_i \Delta \tau), \quad i = 1, \dots, s, \quad j \leq n, \\ e_n(v_j, \Delta \tau) &= \bar{v}_n - u_h(\tau_n), \quad j \leq n.\end{aligned}$$

Then it's easy to derive the following recurrence relations:

$$\begin{aligned}E_{n,i}(v_n, \Delta \tau) &= e^{-c_i \Delta \tau L_h} e_n(v_n, \Delta \tau) + \Delta \tau \sum_{j=1}^{i-1} a_{ij} (-\Delta \tau L_h) (P_h f(\tau_n + c_j \Delta \tau, \\ &\quad U_{n,j}(v_n, \Delta \tau)) - P_h f(\tau_n + c_j \Delta \tau, u(\tau_n + c_j \Delta \tau))) - \delta_{ni},\end{aligned}\quad (25)$$

$$\begin{aligned}e_{n+1}(v_n, \Delta \tau) &= e^{-\Delta \tau L_h} e_n(v_n, \Delta \tau) + \Delta \tau \sum_{i=1}^s b_i (-\Delta \tau L_h) (P_h f(\tau_n + c_i \Delta \tau, \\ &\quad U_{n,i}(v_n, \Delta \tau)) - P_h f(\tau_n + c_i \Delta \tau, u(\tau_n + c_i \Delta \tau))) - \delta_{n+1}.\end{aligned}\quad (26)$$

Next, we present some results on the bounds of the defect terms in (23) and (24). The following Lemmas 4 and 5 have been proved in [27].

Lemma 4 ([27]). *Let $r = 1$ or 2 . Assume the function f satisfies Assumptions 3.1 and 3.2, and the exact solution $u(t)$ fulfills (18a) and (18b) in Assumption 3.3. For $r = 2$, $u(t)$ additionally fulfills (18c), then it holds that*

$$\|\delta_{ni}^{[r]}\|_1 \lesssim (\Delta \tau)^{r+1} \sup_{0 \leq \eta \leq 1} \|f^{(r)}(\tau_n + \eta \Delta \tau, u(\tau_n + \eta \Delta \tau))\|_1 + h, \quad (27a)$$

$$\left\| \sum_{j=0}^{n-1} e^{-j \Delta \tau L_h} \delta_{n-j}^{[r]} \right\|_1 \lesssim (\Delta \tau)^r \sup_{0 \leq t \leq T} \|f^{(r)}(t, u(t))\|_1 + h, \quad (27b)$$

for any $n = 0, \dots, N_T, i = 1, \dots, s$, and the hidden constants are independent of h and $\Delta \tau$.

Lemma 5 ([27]). Suppose the function f satisfies Assumptions 3.1 and 3.2, and the exact solution $u(t)$ fulfills (18a) and (18b) in Assumption 3.3. If the initial value $v_n \in V_h$, then it holds for any $0 \leq n \leq N_T - 1$, $i = 1, \dots, s$,

$$\|E_{n,i}(v_n, \Delta\tau)\|_1 \lesssim \|v_n - u_h(\tau_n)\|_1 + (\Delta\tau)^2 \sup_{0 \leq \eta \leq 1} \|f'(\tau_n + \eta\Delta\tau, u(\tau_n + \eta\Delta\tau))\|_1 + h, \quad (28)$$

for $s \geq 2$, where the hidden constant is independent of h and $\Delta\tau$.

Similar as the definition in (26), we denote $e_n(v_j, \tau)$ as the numerical error at the time τ_n by applying the EIFE method successively for $(n - j)$ steps with the initial value v_j at τ_j and the time step size τ . Then we have the following error estimate for EIFE1 method.

Theorem 1 ([27]). Suppose the function f satisfies Assumptions 3.1 and 3.2, and the exact solution $u(t)$ fulfills (18a) and (18b). There exists a constant $h_0 > 0$ such that if the spatial mesh size $h \leq h_0$, then for the numerical result produced by EIFE1 method with the initial value $v_j \in V_h$, it holds

$$\|e_n(v_j, \Delta\tau)\|_1 \lesssim \|v_j - u_h(\tau_j)\|_1 + \Delta\tau + h, \quad \forall j \leq n \leq N_T, \quad (29)$$

where the hidden constant is independent of h and $\Delta\tau$.

4 | CONVERGENCE ANALYSIS OF THE PARAREAL METHOD

Define $\tilde{j} = \lfloor \frac{j}{M} \rfloor$ for any $0 \leq j \leq NM - 1$. Based on (11), we denote $\tilde{U}_{j,i}(v_j, \Delta t)$ and $\tilde{u}_h^j(v_j, \Delta t)$ respectively as the i th inner stage and RK numerical result computed on the j th fine time interval after advancing for $(j - \tilde{j}M)$ steps by the fine propagator $\mathcal{F}$ with the initial value v_j and the fine time step size Δt . Moreover, we define

$$\begin{aligned} \tilde{E}_{j,i}(v_j, \Delta t) &= \tilde{U}_{j,i}(v_j, \Delta t) - u_h(t_j + c_i \Delta t), \\ \tilde{E}_j(v_j, \Delta t) &= \tilde{u}_h^j(v_j, \Delta t) - u_h(t_j) \end{aligned}$$

as the i th inner stage error and RK numerical errors on the j th fine interval, respectively. For any $1 \leq n \leq N$, based on the Parareal-EIFE method proposed in Algorithm 1, we can derive by recurrence the following result for any $k \geq n$,

$$\begin{aligned} U_h^{n,(k)} - u_h(T_n) &= \mathcal{G}(U_h^{n-1,(k)}; T_{n-1}, T_n) + \mathcal{F}^M(U_h^{n-1,(k-1)}; T_{n-1}, T_n) \\ &\quad - \mathcal{G}(U_h^{n-1,(k-1)}; T_{n-1}, T_n) - \mathcal{H}(u_h(T_{n-1}); T_{n-1}, T_n) \\ &=: \mathbf{I}_1 + \mathbf{I}_2 + \mathbf{I}_3, \end{aligned} \quad (30)$$

where

$$\begin{aligned} \mathbf{I}_1 &= \sum_{j=0}^{nM-1} \Delta t e^{-(nM-1-j)\Delta t L_h} \sum_{i=1}^q b_i(-\Delta t L_h) \left(P_h f(t_j + c_i \Delta t, \tilde{U}_{j,i}(U_h^{j,(k-n+j)}), \Delta t) \right. \\ &\quad \left. - P_h f(t_j + c_i \Delta t, u(t_j + c_i \Delta t)) \right), \\ \mathbf{I}_2 &= \sum_{j=0}^{n-1} \Delta T e^{-(n-1-j)\Delta T L_h} \sum_{i=1}^p b_i(-\Delta T L_h) \left(P_h f(T_j + c_i \Delta T, U_{j,i}(U_h^{j,(k-n+1+j)}), \Delta T) \right. \\ &\quad \left. - P_h f(T_j + c_i \Delta T, U_{j,i}(U_h^{j,(k-n+j)}), \Delta T) \right), \\ \mathbf{I}_3 &= - \sum_{j=0}^{nM-1} e^{-j\Delta t L_h} \delta_{nM-1-j}. \end{aligned}$$

Define $\tilde{l} = \lfloor \frac{l}{M} \rfloor$ and $\underline{l} = \tilde{l}M$ for any $0 \leq l \leq NM - 1$.

Lemma 6. Suppose the function f satisfies Assumptions 3.1 and 3.2, and the exact solution $u(t)$ fulfills (18a) and (18b) in Assumption 3.3. Assume $q \geq 2$. For any $0 \leq l \leq NM - 1$ and $1 \leq i \leq q$, if the initial value $v_l \in V_h$, then it holds

$$\begin{aligned} \|\tilde{E}_{l,i}(v_l, \Delta t)\|_1 &\lesssim \|v_l - u_h(T_l)\|_1 + (\Delta t)^2 \sup_{0 \leq \eta \leq 1} \|f'(t_l + \eta \Delta t, u(t_l + \eta \Delta t))\|_1 \\ &\quad + (\Delta t)^2 \sup_{0 \leq \eta \leq 1} \|f^{(2)}(t_l + \eta \Delta t, u(t_l + \eta \Delta t))\|_1 + h, \end{aligned} \quad (31)$$

where the hidden constant is independent of h and Δt .

Proof. According to the definition of $\psi_{j,i}$ in (22), we have that $\psi_{1,j} = 0$ for $j = 1, \dots, q$ by some direct manipulations since the EIFE method proposed in (9) is consistent. Then the defects $\delta_{li} = \delta_{li}^{[1]}$ for $i = 1, \dots, q$ when $q \geq 2$. With the help of recurrence and (25) and (26), we can obtain the following inequality:

$$\begin{aligned} \|\tilde{E}_{l,i}(v_l, \Delta t)\|_1 &= \left\| e^{-(l-l+c_i)\Delta t L_h} (v_l - u_h(T_l)) + \Delta t \sum_{j=0}^{l-l-1} e^{-(l-l-1-j+c_i)\Delta t L_h} \sum_{i'=1}^q b_{i'}(-\Delta t L_h) \right. \\ &\quad \left(P_h f(t_{l+j} + c_{i'} \Delta t, \tilde{U}_{l+j,i'}(v_l, \Delta t)) - P_h f(t_{l+j} + c_{i'} \Delta t, u(t_{l+j} + c_{i'} \Delta t)) \right) \\ &\quad + \Delta t \sum_{j'=1}^{i-1} a_{ij'}(-\Delta t L_h) \left(P_h f(t_l + c_{j'} \Delta t, \tilde{U}_{l,j'}(v_l, \Delta t)) \right. \\ &\quad \left. - P_h f(t_l + c_{j'} \Delta t, u(t_l + c_{j'} \Delta t)) \right) - \sum_{j=0}^{l-l-1} e^{-(j+c_i)\Delta t L_h} \delta_{l-1-j} - \delta_{li} \Big\|_1 \\ &\leq \Pi_1 + \Pi_2 + \Pi_3 + \Pi_4, \end{aligned} \quad (32)$$

where

$$\begin{aligned} \Pi_1 &= \left\| \Delta t \sum_{j=0}^{l-l-1} e^{-(l-l-1-j+c_i)\Delta t L_h} \sum_{i'=1}^q b_{i'}(-\Delta t L_h) \left(P_h f \left(t_{l+j} + c_{i'} \Delta t, \right. \right. \right. \\ &\quad \left. \left. \tilde{U}_{l+j,i'}(v_l, \Delta t) \right) - P_h f(t_{l+j} + c_{i'} \Delta t, u(t_{l+j} + c_{i'} \Delta t)) \right) \Big\|_1 \\ \Pi_2 &= \left\| \Delta t \sum_{j'=1}^{i-1} a_{ij'}(-\Delta t L_h) \left(P_h f(t_l + c_{j'} \Delta t, \tilde{U}_{l,j'}(v_l, \Delta t)) - P_h f(t_l + c_{j'} \Delta t, u(t_l + c_{j'} \Delta t)) \right) \right\|_1 \\ \Pi_3 &= \|e^{-(l-l+c_i)\Delta t L_h} (v_l - u_h(T_l))\|_1, \quad \Pi_4 = \left\| \sum_{j=0}^{l-l-1} e^{-(j+c_i)\Delta t L_h} \delta_{l-1-j} + \delta_{li} \right\|_1. \end{aligned}$$

Noting (13) and (14) in Lemma 1, we derive that

$$\begin{aligned} \|\Pi_1\|_1 &\lesssim \left\| \Delta t e^{-c_i \Delta t L_h} \sum_{i'=1}^q b_{i'}(-\Delta t L_h) \left(P_h f(t_{l-1} + c_{i'} \Delta t, \tilde{U}_{l-1,i'}(v_l, \Delta t)) - P_h f(t_{l-1} + c_{i'} \Delta t, u(t_{l-1} + c_{i'} \Delta t)) \right) \right\|_1 \\ &\quad + \left\| \Delta t \sum_{j=0}^{l-l-2} e^{-(l-l-1-j+c_i)\Delta t L_h} \sum_{i'=1}^q b_{i'}(-\Delta t L_h) \left(P_h f(t_{l+j} + c_{i'} \Delta t, \right. \right. \\ &\quad \left. \left. \tilde{U}_{l+j,i'}(v_l, \Delta t) \right) - P_h f(t_{l+j} + c_{i'} \Delta t, u(t_{l+j} + c_{i'} \Delta t)) \right) \Big\|_1 \end{aligned} \quad (33)$$

$$\begin{aligned}
&\lesssim (\Delta t)^{\frac{1}{2}} \max_{1 \leq i' \leq q} \|\tilde{E}_{l-1,i'}(v_l, \Delta t)\|_1 + \|e^{-c_i \Delta t L_h}\|_0 \Delta t \sum_{j=0}^{l-l-2} (t_{l-l-j})^{-\frac{1}{2}} \max_{1 \leq i' \leq q} \|\tilde{E}_{l+j,i'}(v_l, \Delta t)\|_1 + h \\
&\lesssim \Delta t \sum_{j=0}^{l-l-1} (t_{l-l-j})^{-\frac{1}{2}} \max_{1 \leq i' \leq q} \|\tilde{E}_{l+j,i'}(v_l, \Delta t)\|_1 + h.
\end{aligned}$$

As for the estimation of the second term, with the help of (13), we have

$$\begin{aligned}
\|\Pi_2\|_1 &\lesssim (\Delta t)^{\frac{1}{2}} \max_{1 \leq j' \leq i-1} \|P_h f(t_l + c_i \Delta t, \tilde{U}_{l,j'}(v_l, \Delta t)) - P_h f(t_l + c_j \Delta t, u(t_l + c_j \Delta t))\|_0 \\
&\lesssim (\Delta t)^{\frac{1}{2}} \max_{1 \leq j' \leq i-1} \|\tilde{U}_{l,j'}(v_l, \Delta t) - u_h(t_l + c_j \Delta t)\|_1 + h.
\end{aligned} \tag{34}$$

For the third term in (32), according to the relation between H^1 and L^2 norms in (15), we have

$$\begin{aligned}
\|\Pi_3\|_1 &= \|e^{-(l-l+c_i)\Delta t L_h}(v_l - u_h(T_l))\|_1 \\
&= \|L_h^{\frac{1}{2}} e^{-(l-l+c_i)\Delta t L_h}(v_l - u_h(T_l))\|_0 \\
&\leq \|e^{-(l-l+c_i)\Delta t L_h}\|_0 \|L_h^{\frac{1}{2}}(v_l - u_h(T_l))\|_0 \\
&\lesssim \|v_l - u_h(T_l)\|_1,
\end{aligned} \tag{35}$$

where, in the derivation of the last inequality, we have used the estimate (12) in Lemma 1, since $l - l + c_i \geq 0$.

Recalling the definition of ψ_i in (22), we can check that $\psi_1(-\Delta L_h) = \psi_2(-\Delta t L_h) = 0$, which implies that $\delta_{l+1} = \delta_{l+1}^{[2]}$ by Lagrangian interpolation theorem for $q \geq 2$. Inserting the conclusion of (27a) (with $r = 1$) and (27b) (with $r = 2$) into Lemma 4, we can obtain that

$$\begin{aligned}
\|\Pi_4\|_1 &\leq \left\| \sum_{j=0}^{l-l-1} e^{-(j+c_i)\Delta t L_h} \delta_{l-1-j}^{[2]} \right\|_1 + \|\delta_{li}^{[1]}\|_1 \\
&\lesssim (\Delta t)^2 \sup_{0 \leq \eta \leq 1} \|f'(t_l + \eta \Delta t, u(t_l + \eta \Delta t))\|_1 + (\Delta t)^2 \sup_{0 \leq \eta \leq 1} \|f^{(2)}(t_l + \eta \Delta t, u(t_l + \eta \Delta t))\|_1 + h,
\end{aligned} \tag{36}$$

where the last inequality is due to the fact that P_h is H^1 -stable. Combining the results (33)–(36) together, we can reformulate (32) as

$$\begin{aligned}
\|\tilde{E}_{li}(v_l, \Delta t)\|_1 &\lesssim \|v_l - u_h(T_l)\|_1 + \Delta t \sum_{j=0}^{l-l-1} (t_{l-l-j})^{-\frac{1}{2}} \max_{1 \leq i' \leq q} \|\tilde{E}_{l+j,i'}(v_l, \Delta t)\|_1 \\
&\quad + \Delta t^{\frac{1}{2}} \max_{1 \leq j' \leq i-1} \|\tilde{E}_{l,j'}(v_l, \Delta t)\|_1 + (\Delta t)^2 \sup_{0 \leq \eta \leq 1} \|f'(t_l + \eta \Delta t, u(t_l + \eta \Delta t))\|_1 \\
&\quad + (\Delta t)^2 \sup_{0 \leq \eta \leq 1} \|f^{(2)}(t_l + \eta \Delta t, u(t_l + \eta \Delta t))\|_1 + h.
\end{aligned}$$

Finally, with the help of Lemma 5 and the discrete Gronwall inequality, (31) is obtained by recursively using the above inequality. ■

Lemma 7. *If the progression*

$$\sum_{k=0}^{\infty} (a_k - b_k) \xi^k \leq 0, \quad \forall \xi \in (0, 1), \tag{37}$$

is held, then $a_k \leq b_k, \forall k \geq 0$.

Proof. Since (37) is held for any $\xi \in (0, 1)$, taking $\xi \rightarrow 0$, we can first derive that $a_0 \leq b_0$. To ensure the inequality (37) is held, then

$$\sum_{k=1}^{\infty} (a_k - b_k) \xi^k = \xi \sum_{k=0}^{\infty} (a_{k+1} - b_{k+1}) \xi^k \leq 0.$$

Similarly, taking $\xi \rightarrow 0$, we can easily obtain that $a_1 \leq b_1$. As so on, we can derive that $a_k \leq b_k$ is held for any $k \geq 0$. ■

Theorem 2. Assume the function f satisfies Assumptions 3.1 and 3.2 and the exact solution $u(t)$ fulfills (18a) and (18b) in Assumption 3.3. Suppose that either the EIFE1 or EIFE2 scheme is selected as the fine propagator $\mathcal{F}$ for the Parareal-EIFE method (i.e., Algorithm 1), and additionally assume $u(t)$ fulfills (18c) if the EIFE2 scheme is chosen. Then for a given n ($1 \leq n \leq N$),

$$\|U_h^{n,(k)} - u_h(T_n)\|_1 \lesssim (\Delta t)^r + h, \quad \forall k \geq n,$$

where $r = 1$ for the EIFE1 fine propagator and $r = 2$ for the EIFE2 fine propagator, and the hidden constant is independent of h and Δt . Especially, when $k \geq N$

$$\|U_h^{n,(k)} - u_h(T_n)\|_1 \lesssim (\Delta t)^r + h, \quad \forall 1 \leq n \leq N.$$

Proof. Let us first derive the estimate of (30). Noting (13) in Lemma 1, we can obtain that

$$\left\| \Delta t L_h^{\frac{1}{2}} \sum_{i=1}^q b_i(-\Delta t L_h) \right\|_0 \lesssim \Delta t^{\frac{1}{2}}, \quad \left\| \sum_{i=1}^q b_i(-\Delta t L_h) \right\|_0 \lesssim 1. \quad (38)$$

With the help of (12) in Lemma 1, we can derive that

$$\begin{aligned} \left\| L_h^{\frac{1}{2}} \Delta t e^{-(nM-1-j)\Delta t L_h} \right\|_0 &= \Delta t ((nM-1-j)\Delta t)^{-\frac{1}{2}} \left\| L_h^{\frac{1}{2}} ((nM-1-j)\Delta t)^{\frac{1}{2}} e^{-(nM-1-j)\Delta t L_h} \right\|_0 \\ &\lesssim \Delta t (t_{nM-1-j})^{-\frac{1}{2}}. \end{aligned} \quad (39)$$

Then combining (38) and (39), we have

$$\begin{aligned} \|I_1\|_1 &\lesssim \left\| \Delta t L_h^{\frac{1}{2}} \sum_{i=1}^q b_i(-\Delta t L_h) \left(P_h f(t_{nM-1} + c_i \Delta t, \tilde{U}_{nM-1,i}(U_h^{n-1,(k-1)}, \Delta t)) \right. \right. \\ &\quad \left. \left. - P_h f(t_{nM-1} + c_i \Delta t, u_h(t_{nM-1} + c_i \Delta t)) \right) \right\|_0 \\ &\quad + \sum_{j=0}^{nM-2} \left\| L_h^{\frac{1}{2}} \Delta t e^{-(nM-1-j)\Delta t L_h} \sum_{i=1}^q b_i(-\Delta t L_h) \left(P_h f(t_j + c_i \Delta t, \right. \right. \\ &\quad \left. \left. \tilde{U}_{j,i}(U_h^{\tilde{j},(k-n+\tilde{j})}, \Delta t)) - P_h f(t_j + c_i \Delta t, u_h(t_j + c_i \Delta t)) \right) \right\|_0 \\ &\lesssim \Delta t \sum_{j=0}^{nM-1} (t_{nM-j})^{-\frac{1}{2}} \max_{1 \leq i \leq q} \left\| \tilde{E}_{j,i}(U_h^{\tilde{j},(k-n+\tilde{j})}, \Delta t) \right\|_1 + h, \end{aligned} \quad (40)$$

where we use the fact that f is locally-Lipschitz continuous in the last inequality. In the same way,

$$\begin{aligned}
\|I_2\|_1 &\lesssim \Delta T \left\| \sum_{i=1}^p b_i(-\Delta TL_h) \left(P_h f(T_{n-1} + c_i \Delta T, U_{n-1,i}(U_h^{n-1,(k)}, \Delta T)) \right. \right. \\
&\quad \left. \left. - P_h f(T_{n-1} + c_i \Delta T, U_{n-1,i}(U_h^{n-1,(k-1)}, \Delta T)) \right) \right\|_1 \\
&\quad + \Delta T \left\| \sum_{j=0}^{n-2} e^{-(n-1-j)\Delta TL_h} \sum_{i=1}^p b_i(-\Delta TL_h) \left(P_h f \left(T_j + c_i \Delta T, \right. \right. \right. \\
&\quad \left. \left. U_{j,i}(U_h^{j,(k-n+1+j)}, \Delta T) \right) - P_h f(T_j + c_i \Delta T, U_{j,i}(U_h^{j,(k-n+j)}, \Delta T)) \right) \right\|_1 \\
&\lesssim \max_{1 \leq i \leq p} (\Delta T)^{\frac{1}{2}} \left\| P_h f(T_{n-1} + c_i \Delta T, U_{n-1,i}(U_h^{n-1,(k)}, \Delta T)) \right. \\
&\quad \left. - P_h f(T_{n-1} + c_i \Delta T, U_{n-1,i}(U_h^{n-1,(k-1)}, \Delta T)) \right\|_0 \\
&\quad + \Delta T \sum_{j=0}^{n-2} T_{n-1-j}^{-\frac{1}{2}} \max_{1 \leq i \leq p} \left\| P_h f(T_j + c_i \Delta T, U_{j,i}(U_h^{j,(k-n+1+j)}, \Delta T)) \right. \\
&\quad \left. - P_h f(T_j + c_i \Delta T, U_{j,i}(U_h^{j,(k-n+j)}, \Delta T)) \right\|_0 \\
&\lesssim \Delta T \sum_{j=0}^{n-1} (T_{n-j})^{-\frac{1}{2}} \max_{1 \leq i \leq p} \|U_{j,i}(U_h^{j,(k-n+1+j)}, \Delta T) - U_{j,i}(U_h^{j,(k-n+j)}, \Delta T)\|_1.
\end{aligned} \tag{41}$$

Owing to the numerical scheme (9), we can obtain that

$$\begin{aligned}
&\|U_{j,i}(U_h^{j,(k-n+1+j)}, \Delta T) - U_{j,i}(U_h^{j,(k-n+j)}, \Delta T)\|_1 \\
&= \left\| e^{-c_i \Delta TL_h} \left(U_h^{j,(k-n+1+j)} - U_h^{j,(k-n+j)} \right) + \Delta T \sum_{s=1}^{i-1} a_{is}(-\Delta TL_h) \left(P_h f \left(T_j + c_s \Delta T, \right. \right. \right. \\
&\quad \left. \left. U_{j,s}(U_h^{j,(k-n+j+1)}, \Delta T) \right) - P_h f(T_j + c_s \Delta T, U_{j,s}(U_h^{j,(k-n+j)}, \Delta T)) \right) \right\|_1 \\
&\lesssim \|U_h^{j,(k-n+1+j)} - U_h^{j,(k-n+j)}\|_1 + (\Delta T)^{\frac{1}{2}} \max_{2 \leq s \leq i-1} \|U_{j,s}(U_h^{j,(k-n+j+1)}, \Delta T) - U_{j,s}(U_h^{j,(k-n+j)}, \Delta T)\|_1.
\end{aligned}$$

By applying the above inequality repeatedly, we obtain

$$\max_{1 \leq i \leq p} \|U_{j,i}(U_h^{j,(k-n+1+j)}, \Delta T) - U_{j,i}(U_h^{j,(k-n+j)}, \Delta T)\|_1 \lesssim \|U_h^{j,(k-n+1+j)} - u_h(T_j)\|_1 + \|U_h^{j,(k-n+j)} - u_h(T_j)\|_1. \tag{42}$$

Inserting (42) into (41) gives us

$$\|I_2\|_1 \lesssim \Delta T \sum_{j=0}^{n-1} (T_{n-j})^{-\frac{1}{2}} \left(\|U_h^{j,(k-n+1+j)} - u_h(T_j)\|_1 + \|U_h^{j,(k-n+j)} - u_h(T_j)\|_1 \right). \tag{43}$$

Recalling the result of (27b) in Lemma 4 and semi-discrete error estimate (3), we can derive that

$$\|I_3\|_1 \lesssim (\Delta t)^r \sup_{0 \leq t \leq T_n} \|f^{(r)}(t, u(t))\|_1 + h, \tag{44}$$

where $r = 1$ for the EIFE1 fine propagator since $\delta_j = \delta_j^{[1]}$ and $r = 2$ for the EIFE2 fine propagator since $\delta_j = \delta_j^{[2]}$.

Combining (40)–(44), the estimate of $\|U_h^{n,(k)} - u_h(T_n)\|_1$ becomes

$$\begin{aligned} \|U_h^{n,(k)} - u_h(T_n)\|_1 &\lesssim \Delta t \sum_{j=0}^{nM-1} (t_{nM-j})^{-\frac{1}{2}} \max_{1 \leq i \leq q} \|\tilde{E}_{j,i}(U_h^{\tilde{j},(k-n+j)}, \Delta t)\|_1 + h \\ &\quad + \Delta T \sum_{j=0}^{n-1} (T_{n-j})^{-\frac{1}{2}} \left(\|U_h^{j,(k-n+1+j)} - u_h(T_j)\|_1 + \|U_h^{j,(k-n+j)} - u_h(T_j)\|_1 \right) \\ &\quad + (\Delta t)^r \sup_{0 \leq t \leq T_n} \|f^{(r)}(t, u(t))\|_1. \end{aligned}$$

According to Lemma 6, (45) can be converted to

$$\|U_h^{n,(k)} - u_h(T_n)\|_1 \lesssim \Delta T \sum_{j=0}^{n-1} T_{n-j}^{-\frac{1}{2}} \left(\|U_h^{j,(k-n+1+j)} - u_h(T_j)\|_1 + \|U_h^{j,(k-n+j)} - u_h(T_j)\|_1 \right) + h + (\Delta t)^r, \quad (45)$$

with $r = 1$ for the EIFE1 fine propagator and $r = 2$ for the EIFE2 fine propagator. Define

$$\rho_n(\xi) = \sum_{k=n}^{\infty} \|U_h^{n,(k)} - u_h(T_n)\|_1 \xi^k, \quad \xi \in (0, 1).$$

Then (45) can be reformulated as

$$\rho_n(\xi) \lesssim \left(\frac{\Delta T}{\xi} + \Delta T \right) \sum_{j=0}^{n-1} (T_{n-j})^{-\frac{1}{2}} \rho_j(\xi) \xi^{n-j} + \sum_{k=n}^{\infty} (h + (\Delta t)^r) \xi^k. \quad (46)$$

Combining with the discrete Gronwall inequality (Theorem 6.1 with $\alpha = \frac{1}{2}$ in [10]),

$$\sum_{k=n}^{\infty} \|U_h^{n,(k)} - u_h(T_n)\|_1 \xi^k \lesssim \sum_{k=n}^{\infty} (h + (\Delta t)^r) \xi^k.$$

According to Lemma 7, we can obtain that

$$\|U_h^{n,(k)} - u_h(T_n)\|_1 \lesssim h + (\Delta t)^r, \quad \forall k \geq n.$$

Especially, when $n = N$, we have the uniform error estimation

$$\|U_h^{n,(k)} - u_h(T_n)\|_1 \lesssim h + (\Delta t)^r, \quad \forall k \geq N,$$

which completes the proof. $\blacksquare$

Remark 1. In reality, for solving large-scale problems, the parareal algorithm can achieve excellent accuracy even when the number of iteration is much smaller than the number of processors, that is, $k \ll N$. Moreover, the number of processors cannot be too large, since the cost of communication and sequential correction will increase along with the number of processors.

By using the triangle inequality

$$\|U_h^n - u(T_n)\|_1 \leq \|U_h^n - u_h(T_n)\|_1 + \|u_h(T_n) - u(T_n)\|_1,$$

we finally obtain the following result by combining Lemma 3 and Theorem 2.

Theorem 3. Under the conditions of Lemma 3 and Theorem 2, it holds for the Parareal-EIFE method that

$$\|U_h^{n,(k)} - u(T_n)\|_1 \lesssim (\Delta t)^r + h, \quad \forall k \geq N, \quad 1 \leq n \leq N,$$

where $r = 1$ for the EIFE1 fine propagator and $r = 2$ for the EIFE2 fine propagator, and the hidden constant is independent of h and Δt .

Remark 2. For the model problem (1) with the periodic boundary condition, we can convert the original equation into a new one by transforming $v(t, x) = e^{-\alpha t} u(t, x)$ with a constant $\alpha > 0$ and obtaining a coercive bilinear form. In addition, the spatial mesh also can be relaxed to quasi-uniform ones while similar theoretical analysis still remains valid.

Remark 3. Convergence analysis of higher-order Parareal-EIFE schemes with $q \geq 3$ for the fine propagator $\mathcal{F}$ remains to be explored in the future since $\|e_n\|_1$ and $\|E_{ni}\|_1$ will couple together and make the error estimation very challenging.

5 | NUMERICAL EXPERIMENTS AND APPLICATIONS

In this section, some numerical examples and applications are presented to illustrate the performance of the proposed Parareal-EIFE method (Algorithm 1) and verify the convergent results derived in Section 4. FFT-based implementation is used for both the EIFE fine and coarse propagators [27]. All numerical experiments are done on an Intel Gold 5117, 2.00 GHz CPU server with 128 GB memory. Specially, we choose the interpolation node $c_2 = \frac{1}{2}$ for the EIFE2 scheme. For brevity, we also denote the method with the EIFE1 coarse propagator and the EIFE2 fine propagator as Parareal-EIFE12 method. Similarly, we denote Parareal-EIFE11 and Parareal-EIFE22 as the method with the EIFE1 or EIFE2 schemes as both coarse and fine propagators, respectively.

5.1 | Convergence tests

We first verify the error estimates obtained for the Parareal-EIFE method in Theorem 3, and the numerical errors are all evaluated at the terminal time T .

Example 5.1. In this example, we consider the following two-dimensional nonlinear reaction-diffusion problem with the homogeneous Dirichlet boundary condition:

$$\begin{cases} u_t = \frac{1}{2} \Delta u - \frac{1}{2} \pi^2 u + \frac{1}{2} \pi^2 e^{-\pi^2 t} \sin(\pi x) \sin(\pi y), & (x, y) \in \Omega, \quad 0 \leq t \leq T \\ u(0, x, y) = (\sin(\pi x) - 1) \sin(\pi y), & (x, y) \in \Omega, \end{cases}$$

where $\Omega = \left[\frac{1}{2}, \frac{5}{2}\right] \times [0, 1]$ and the terminal time $T = 1$. The exact solution is given by

$$u(t, x, y) = e^{-\pi^2 t} (\sin(\pi x) - 1) \sin(\pi y).$$

For testing the spatial accuracy, we run the Parareal-EIFE22 scheme with fixed temporal partition $N \times M = 4 \times 256$ (i.e., $\Delta T = T/N = 1/4$, $\Delta t = \Delta T/M = 1/1024$) and uniformly refined spatial grids with $N_x \times N_y = 64 \times 32, 128 \times 64, 256 \times 128$ and 512×256 , respectively. For testing the temporal accuracy, we fix the spatial grid with $N_x \times N_y = 2048 \times 1024$, and the temporal partitions are $N \times M = 4 \times 8, 4 \times 16, 4 \times 32, 4 \times 64$ for Parareal-EIFE11 method and $N \times M = 4 \times 4, 4 \times 8, 4 \times 16, 4 \times 32$ for Parareal-EIFE12 and Parareal-EIFE22 method, respectively. Specifically, in order to verify that the temporal accuracy of the Parareal-EIFE method can reach to that of the EIFE method with the small time step size when the iteration number (for the prediction-correction procedure) $k \geq N$ in Algorithm 1, we fix the number of iterations be 4 (i.e., terminate the iteration when $k = 4$) since $N = 4$ in these cases. All numerical results are reported in Table 1, including the numerical errors measured in L^2 and H^1 norms and the corresponding convergence rates. We also plot the numerical

TABLE 1 Numerical results on the solution errors measured in the L^2 and H^1 norms and corresponding convergence rates for the Parareal-EIFE11, Parareal-EIFE12 and Parareal-EIFE22 schemes in Example 5.1.

Spatial accuracy tests for Parareal-EIFE22					
$N \times M$	$N_x \times N_y$	$\ U_h^{n,(4)} - u(T_n)\ _0$	CR	$\ U_h^{n,(4)} - u(T_n)\ _1$	CR
4×256	64×32	$5.2210\text{e-}07$	-	$4.5560\text{e-}06$	-
4×256	128×64	$1.3030\text{e-}07$	2.00	$2.1471\text{e-}06$	1.09
4×256	256×128	$3.1880\text{e-}08$	2.03	$1.0564\text{e-}06$	1.02
4×256	512×256	$7.2463\text{e-}09$	2.14	$5.2604\text{e-}07$	1.01
Temporal accuracy tests for EIFE1					
32	2048×1024	$1.6807\text{e-}05$	-	$6.1655\text{e-}05$	-
64	2048×1024	$9.2841\text{e-}06$	0.86	$3.4016\text{e-}05$	0.93
128	2048×1024	$4.8688\text{e-}06$	0.93	$1.7831\text{e-}05$	0.93
256	2048×1024	$2.4920\text{e-}06$	0.97	$9.1250\text{e-}06$	0.97
Temporal accuracy tests for EIFE2					
16	2048×1024	$8.5333\text{e-}06$	-	$3.1262\text{e-}05$	-
32	2048×1024	$1.6087\text{e-}06$	2.41	$5.8940\text{e-}06$	2.41
64	2048×1024	$3.5949\text{e-}07$	2.16	$1.3234\text{e-}06$	2.15
128	2048×1024	$8.4932\text{e-}08$	2.08	$3.3799\text{e-}07$	1.97
Temporal accuracy tests for Parareal-EIFE11					
4×8	2048×1024	$1.6807\text{e-}05$	-	$6.1655\text{e-}05$	-
4×16	2048×1024	$9.2841\text{e-}06$	0.86	$3.4016\text{e-}05$	0.93
4×32	2048×1024	$4.8688\text{e-}06$	0.93	$1.7831\text{e-}05$	0.93
4×64	2048×1024	$2.4920\text{e-}06$	0.97	$9.1250\text{e-}06$	0.97
Temporal accuracy tests for Parareal-EIFE12					
4×4	2048×1024	$8.5333\text{e-}06$	-	$3.1262\text{e-}05$	-
4×8	2048×1024	$1.6087\text{e-}06$	2.41	$5.8940\text{e-}06$	2.41
4×16	2048×1024	$3.5949\text{e-}07$	2.16	$1.3234\text{e-}06$	2.15
4×32	2048×1024	$8.4932\text{e-}08$	2.08	$3.3799\text{e-}07$	1.97
Temporal accuracy tests for Parareal-EIFE22					
4×4	2048×1024	$8.5333\text{e-}06$	-	$3.1262\text{e-}05$	-
4×8	2048×1024	$1.6087\text{e-}06$	2.41	$5.8940\text{e-}06$	2.41
4×16	2048×1024	$3.5949\text{e-}07$	2.16	$1.3234\text{e-}06$	2.15
4×32	2048×1024	$8.4932\text{e-}08$	2.08	$3.3799\text{e-}07$	1.97

errors at times $t = 0.25, 0.5, 1$ in Figure 1, produced by the Parareal-EIFE22 method with spatial meshes $N_x \times N_y = 2048 \times 1024$ and temporal partitions $N \times M = 4 \times 32$. We observe the second-order spatial convergence with respect to L^2 norm and the first-order convergence in H^1 norm as expected. It is also easy to see the first-order temporal convergence for EIFE1 and Parareal-EIFE11 and the second-order temporal convergence for EIFE2, Parareal-EIFE12 and Parareal-EIFE22. Furthermore, the accuracy of Parareal-EIFE method can reach to that of applying EIFE method sequentially with the small time step size as expected.

Table 2 reports the average running (seconds) per iteration for the Parareal-EIFE22 scheme and corresponding growth factors along the refinement of the spatial mesh. All tests are run on 4 cores in parallel with the fixed temporal partition $N \times M = 4 \times 16$ and $k = 4$. The uniform spatial meshes are $N_x \times N_y = 256 \times 128, 512 \times 256, 1024 \times 512$ and 2048×1024 . The results clearly show that the

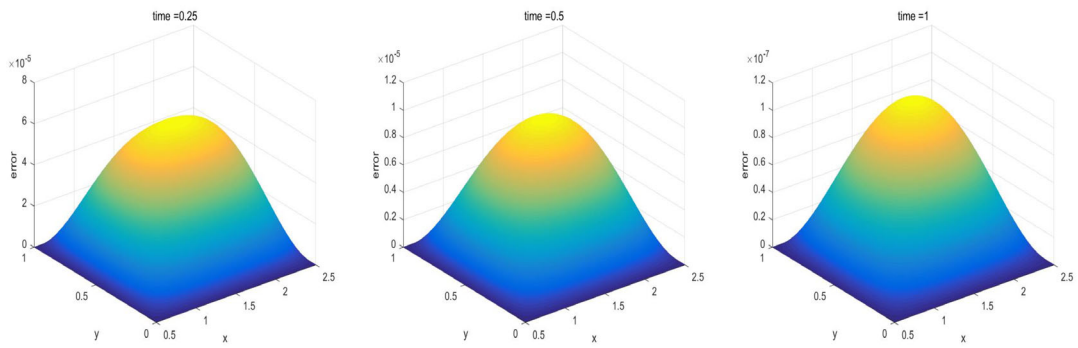


FIGURE 1 Plots of the numerical errors at times $t = 0.25, 0.5, 1$ produced by the Parareal-EIFE22 scheme for Example 5.1. Spatial meshes $N_x \times N_y = 2048 \times 1024$ and temporal partitions $N \times M = 4 \times 32$.

TABLE 2 The average running times (seconds) per iteration under different spatial meshes and corresponding growth factors with respect to the number of mesh nodes for the Parareal-EIFE22 scheme in Example 5.1.

$N_x \times N_y$	$N \times M$	Ave. runtime/iteration	Growth factor
256×128	4×16	29.07	-
512×256	4×16	113.42	0.98
1024×512	4×16	445.54	0.99
2048×1024	4×16	1906.86	1.05

TABLE 3 The running times (seconds) and numerical errors with respect to L^2 and H^1 norms of the Parareal-EIFE and EIFE methods in Example 5.1.

Method	L^2 -error	H^1 -error	Running time
EIFE1	2.4920e-06	9.1250e-06	6848.47
EIFE2	2.0136e-08	1.5087e-07	9315.60
Parareal-EIFE11	2.4923e-06	9.1262e-06	3031.68
Parareal-EIFE12	1.9375e-08	1.4952e-07	4284.76
Parareal-EIFE22	2.0136e-08	1.5087e-07	4938.36

computational cost grows almost linearly along with the number of mesh nodes, which matches well with the property of FFT implementation.

Furthermore, we also compare the performance of the Parareal-EIFE method and the EIFE method. The spatial grid with $N_x \times N_y = 2048 \times 1024$ and the temporal partition with $N \times M = 32 \times 8$ are used. We run the Parareal-EIFE code in parallel on 8 cores with $k = 6$ and the EIFE code sequentially on 1 core. Results reported in Table 3 show that the accuracy of the Parareal-EIFE method can reach that of applying the same EIFE method with the fine step size and the Parareal-EIFE schemes achieve a speedup of roughly 2. Moreover, Figure 2 presents the evolutions (almost monotonically decreasing) of the L^2 and H^1 errors of the numerical solutions produced by the Parareal-EIFE method along with the iteration. We observe that Parareal-EIFE11 needs 4 iterations and Parareal-EIFE12 needs 6 iterations to be practically convergent, and Parareal-EIFE22 only needs about 2 iterations, which are much smaller than the number of coarse time subintervals (i.e., $N = 32$), thus all of them are very efficient due to the fast convergence.

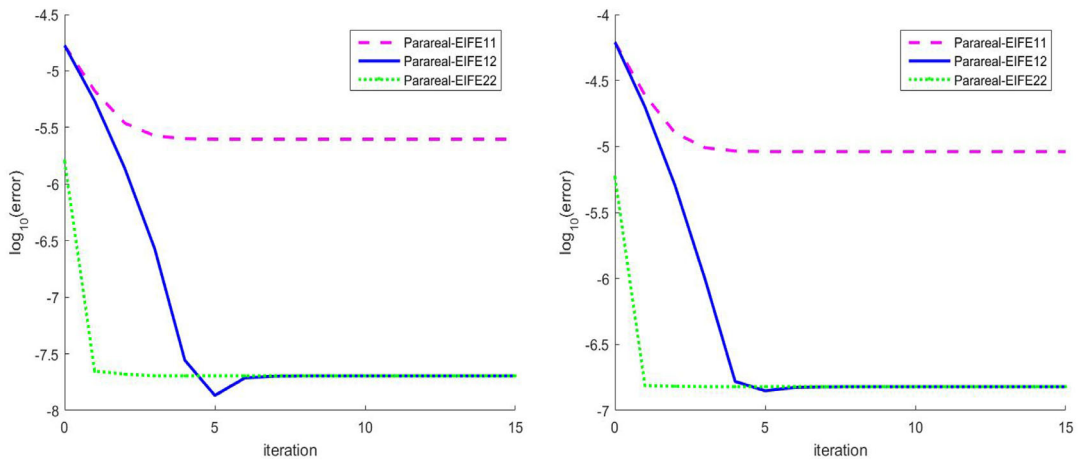


FIGURE 2 The evolutions of L^2 -errors (left) and H^1 -errors (right) of the numerical solutions produced by the Parareal-EIFE11, Parareal-EIFE12 and Parareal-EIFE22 schemes along with the iteration in Example 5.1.

TABLE 4 Numerical results on the solution errors measured in the L^2 and H^1 norms and corresponding convergence rates for the Parareal-EIFE11, Parareal-EIFE12 and Parareal-EIFE22 schemes in Example 5.2.

$N \times M$	$N_x \times N_y \times N_z$	$\ U_h^{n,(4)} - u(T_n)\ _0$	CR	$\ U_h^{n,(4)} - u(T_n)\ _1$	CR
Temporal accuracy tests for EIFE1					
16	$1024 \times 64 \times 64$	2.4126e-04	-	6.7800e-02	-
32	$1024 \times 64 \times 64$	7.3576e-05	1.71	2.1500e-02	1.66
64	$1024 \times 64 \times 64$	2.9275e-05	1.33	8.6000e-03	1.32
128	$1024 \times 64 \times 64$	1.2500e-05	1.23	3.8000e-03	1.18
Temporal accuracy tests for EIFE2					
16	$1024 \times 64 \times 64$	9.5405e-04	-	4.6100e-02	-
32	$1024 \times 64 \times 64$	3.0619e-05	4.95	4.7000e-03	3.29
64	$1024 \times 64 \times 64$	4.5661e-06	2.75	9.3473e-04	2.33
128	$1024 \times 64 \times 64$	9.4147e-07	2.28	2.1521e-04	2.12
Temporal accuracy tests for Parareal-EIFE11					
4×4	$1024 \times 64 \times 64$	2.4126e-04	-	6.7800e-02	-
4×8	$1024 \times 64 \times 64$	7.3576e-05	1.71	2.1500e-02	1.66
4×16	$1024 \times 64 \times 64$	2.9275e-05	1.33	8.6000e-03	1.32
4×32	$1024 \times 64 \times 64$	1.2500e-05	1.23	3.8000e-03	1.18
Temporal accuracy tests for Parareal-EIFE12					
4×4	$1024 \times 64 \times 64$	9.5405e-04	-	4.6100e-02	-
4×8	$1024 \times 64 \times 64$	3.0619e-05	4.95	4.7000e-03	3.29
4×16	$1024 \times 64 \times 64$	4.5661e-06	2.75	9.3473e-04	2.33
4×32	$1024 \times 64 \times 64$	9.4147e-07	2.28	2.1521e-04	2.12
Temporal accuracy tests for Parareal-EIFE22					
4×4	$1024 \times 64 \times 64$	9.5405e-04	-	4.6100e-02	-
4×8	$1024 \times 64 \times 64$	3.0619e-05	4.95	4.7000e-03	3.29
4×16	$1024 \times 64 \times 64$	4.5661e-06	2.75	9.3473e-04	2.33
4×32	$1024 \times 64 \times 64$	9.4147e-07	2.28	2.1521e-04	2.12

Example 5.2. In this example, we consider the traveling wave problem governed by the following 3D Allen–Cahn equation with double-well potential function:

$$\begin{cases} u_t = \Delta u - \frac{1}{\epsilon^2}(u^3 - u), & (x, y, z) \in \Omega, \quad 0 \leq t \leq T, \\ u(0, x, y, z) = \frac{1}{2} \left(1 - \tanh\left(\frac{x}{2\sqrt{2}\epsilon}\right) \right), & (x, y, z) \in \Omega, \end{cases}$$

where $\Omega = [0, \sqrt{2}] \times [0, \frac{1}{8}] \times [0, \frac{1}{8}]$ and $\epsilon = 0.05$. The exact solution is given by

$$u(t, x, y, z) = \frac{1}{2} \left(1 - \tanh\left(\frac{x - st}{2\sqrt{2}\epsilon}\right) \right),$$

where $s = \frac{3}{\sqrt{2}\epsilon}$, and the Dirichlet boundary condition is correspondingly imposed, which is clearly nonhomogeneous. The terminal time is taken to be $T = \frac{3\sqrt{2}\epsilon}{5}$.

To test the temporal accuracy, we fix the uniform spatial meshes with $N_x \times N_y \times N_z = 1024 \times 64 \times 64$ and set the temporal partitions $N \times M = 4 \times 4, 4 \times 8, 4 \times 16, 4 \times 32$ respectively. We again fix the number of iterations to be 4 for all Parareal-EIFE schemes since $N = 4$ in these cases. All numerical results are shown in Table 4, including the numerical errors measured in L^2 and H^1 norms and the corresponding rates. We also plot the numerical solutions and corresponding errors in the cross section $z = 1/16$ at times $t = 0.25T, 0.5T, T$ in Figure 3 for the Parareal-EIFE22 method with spatial meshes $N_x \times N_y \times N_z = 1024 \times 64 \times 64$ and temporal partitions $N \times M = 4 \times 32$. We still observe the first-order temporal convergence for EIFE1 and Parareal-EIFE11 and the second-order for EIFE2,

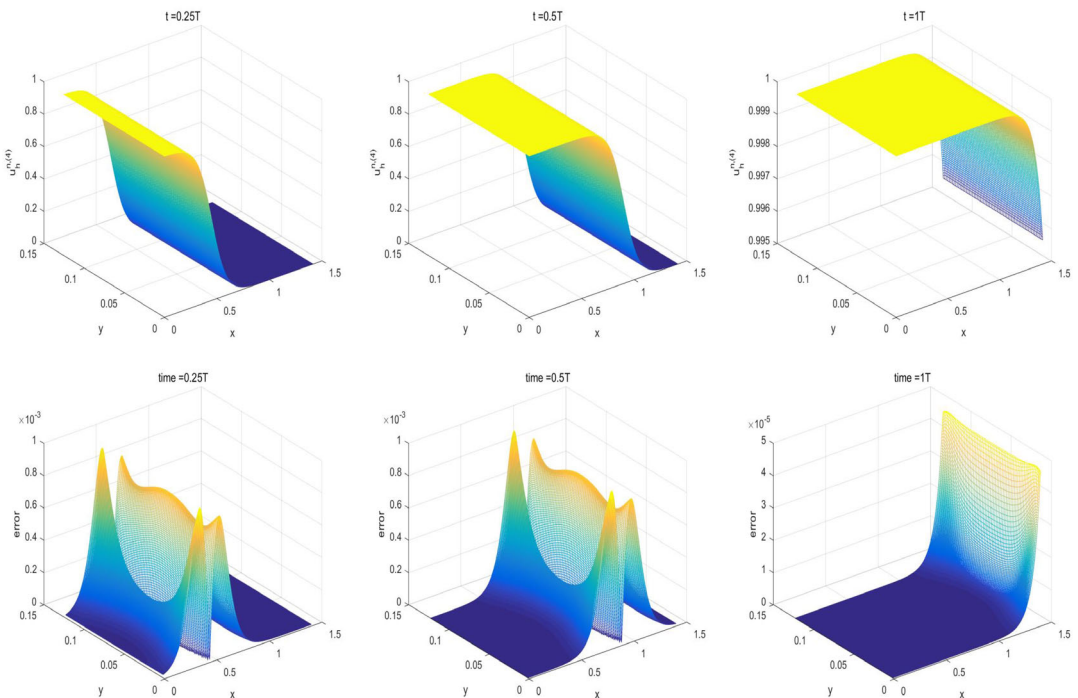


FIGURE 3 Plots of the numerical solution (top) and corresponding error (bottom) in the cross section $z = 1/16$ at times $t = 0.25T, 0.5T, T$ for Example 5.2. Spatial meshes $N_x \times N_y \times N_z = 1024 \times 64 \times 64$ and temporal partitions $N \times M = 4 \times 32$.

TABLE 5 The running time (seconds) and numerical error with respect to L^2 and H^1 norms of Parareal-EIFE and EIFE methods in Example 5.2.

Method	L^2 -error	H^1 -error	Running time
EIFE1	4.9745e-06	1.1000e-03	58799.98
EIFE2	2.8683e-07	5.1049e-05	59716.50
Parareal-EIFE11	4.9745e-06	1.1000e-03	10794.35
Parareal-EIFE12	3.1429e-07	5.1252e-05	11120.49
Parareal-EIFE22	2.8683e-07	5.1049e-05	13959.38

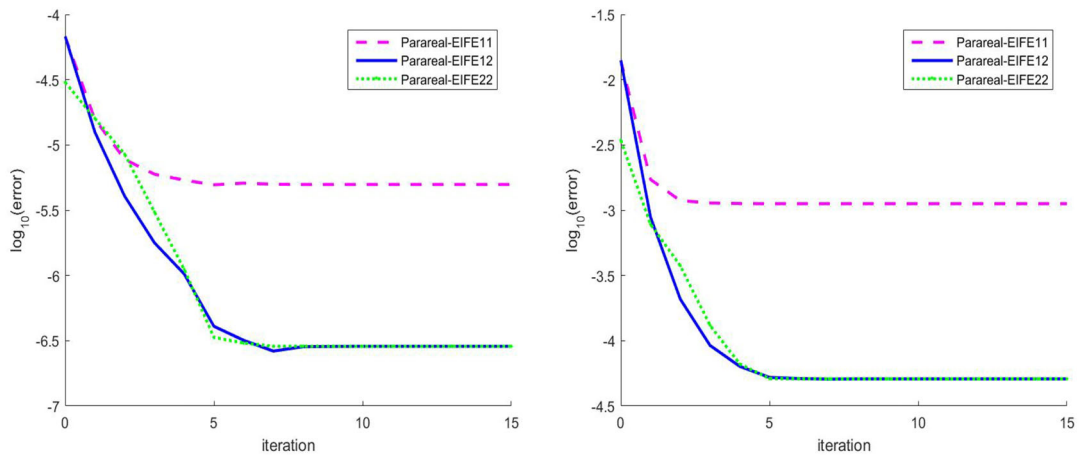


FIGURE 4 The evolutions of L^2 -errors (left) and H^1 -errors (right) of the numerical solutions produced by the Parareal-EIFE11, Parareal-EIFE12 and Parareal-EIFE22 schemes along with the iteration in Example 5.2.

Parareal-EIFE12 and Parareal-EIFE22. Consistent with the theoretical results derived in Theorem 3, the accuracy of the Parareal-EIFE method can reach to that of applying the EIFE method with the small time step size.

Next we compare the performance of the Parareal-EIFE and EIFE methods. The spatial grid with $N_x \times N_y \times N_z = 512 \times 32 \times 32$ and the temporal partition with $N \times M = 32 \times 8$ are taken. We run the Parareal-EIFE code in parallel on 16 cores and fix the number of iterations to be 10. Results reported in Table 5 show again that the Parareal-EIFE schemes can achieve the accuracy of the EIFE ones with the fine step size, but with a speedup of roughly 5. Additionally, the evolutions of the L^2 and H^1 errors of the numerical solutions produced by the Parareal-EIFE method along with the iteration are shown in Figure 4, where the tests are run on 8 cores in parallel. We observe that Parareal-EIFE11, Parareal-EIFE12 and Parareal-EIFE22 need 5 iterations, 8 iterations and 6 iterations to be practically convergent respectively, which are again much smaller than the number of coarse time subintervals (i.e., $N = 32$).

5.2 | 3D grain coarsening simulations

We now illustrate the performance of the proposed Parareal-EIFE method through numerical simulation of the 3D grain coarsening process. In particular, we only simulate Parareal-EIFE22 scheme to obtain 3D phase structures for its higher-order accuracy.

Example 5.3. We consider the grain coarsening process governed by the following 3D Allen–Cahn equation with Flory–Huggins potential function:

$$u_t = \epsilon^2 \Delta u + \frac{\theta}{2} \ln \frac{1-u}{1+u} + \theta_c u, \quad (x, y, z) \in \Omega, \quad 0 < t \leq T,$$

where $\Omega = [0, 1]^3$. The initial data is generated by random numbers on each mesh node ranging from -0.9 to 0.9 , and the periodic boundary condition is imposed. This equation

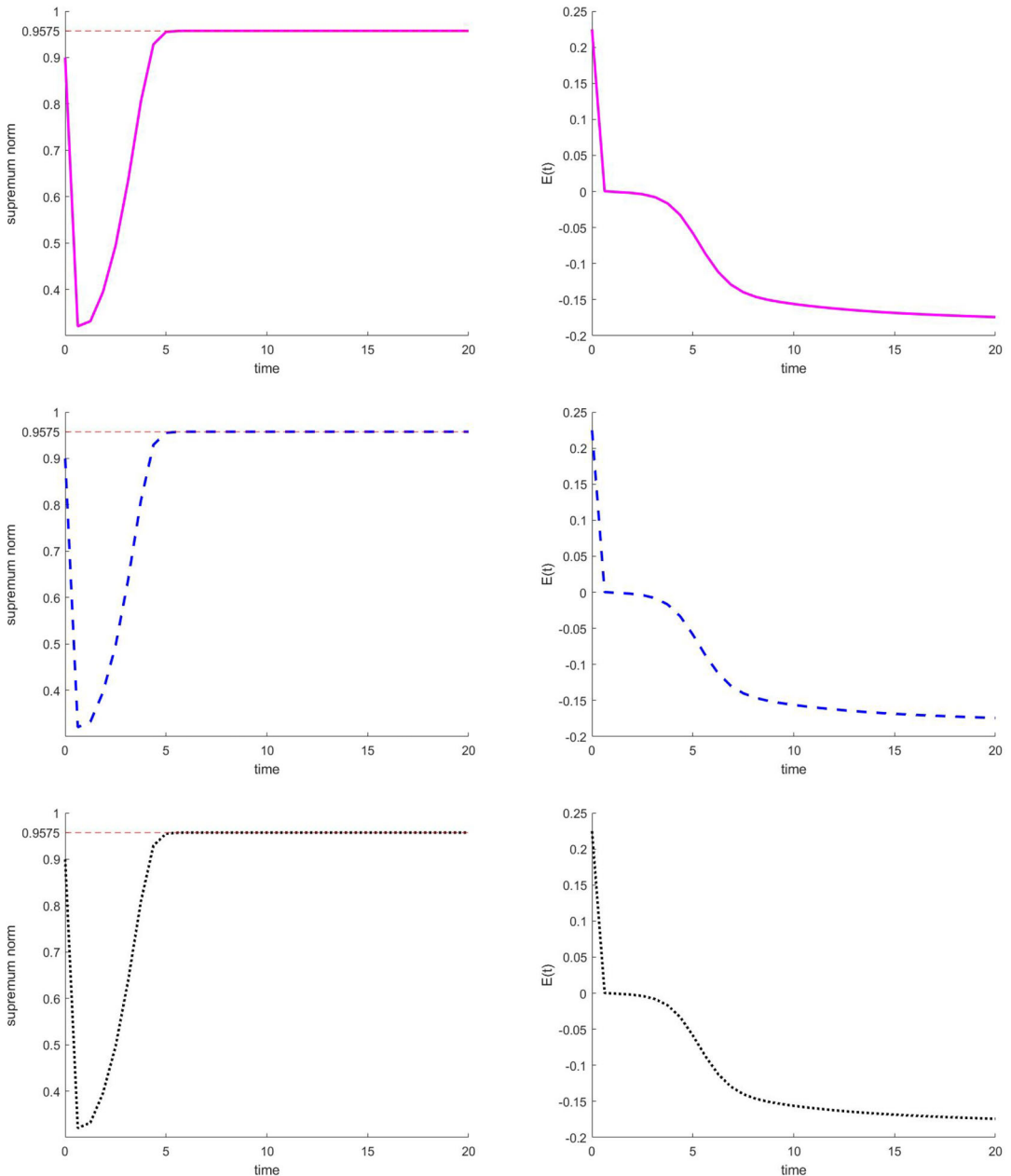


FIGURE 5 The evolutions of supremum norms (left column) and energies (right column) of the numerical solutions produced by Parareal-EIFE11 (top), Parareal-EIFE12 (middle), Parareal-EIFE22 (bottom) schemes for Example 5.3.

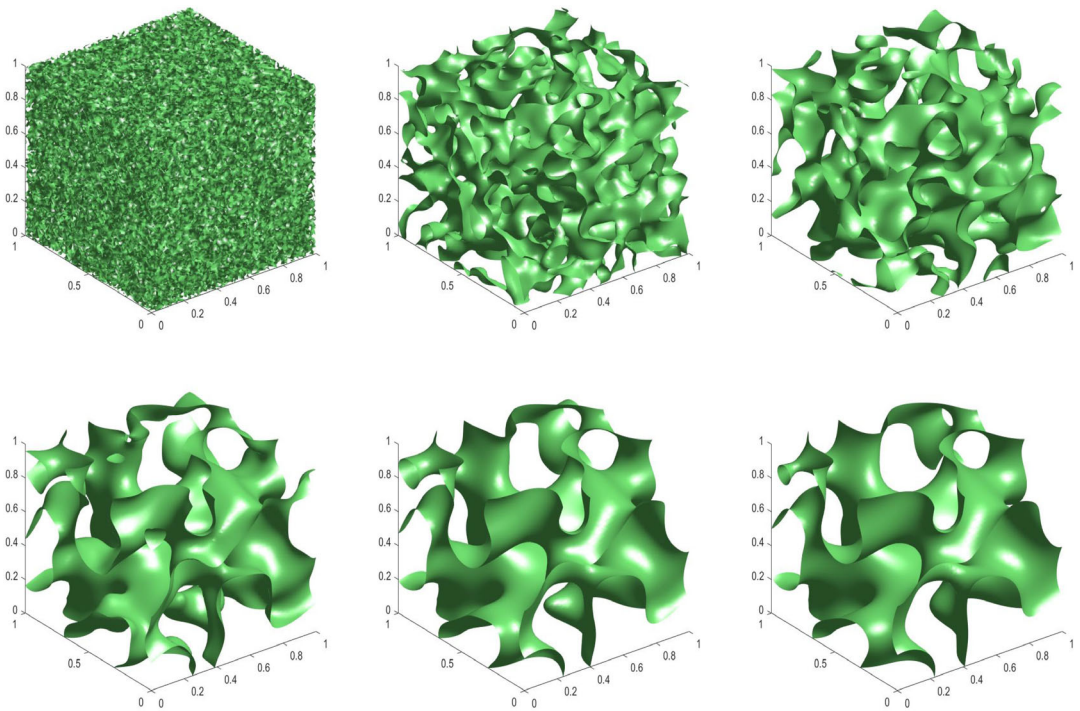


FIGURE 6 Simulated 3D phase structures at $t = 0, 2.5, 5, 10, 15, 20$ (from left to right and top to bottom) produced by the Parareal-EIFE22 scheme for Example 5.3.

can be regarded as the L^2 gradient flow of the following energy function

$$E(u) = \int_{\Omega} \frac{\theta}{2} ((1+u) \ln(1+u) + (1-u) \ln(1-u)) - \frac{\theta_c}{2} u^2 + \frac{\epsilon^2}{2} |\nabla u|^2 dx$$

and thus the energy monotonically decays along the time. We set the interface thickness coefficient $\epsilon = 0.01$ and Flory–Huggins potential parameters $\theta_c = 1.6, \theta = 0.8$.

This test problem satisfies the maximum bound principle with a maximum bound value of $\gamma \approx 0.9575$ [9, 32], that is, $|u(t, x)| \leq \gamma$ for all $x \in \Omega$ and $t \geq 0$. We run simulations until $T = 20$ using the spatial grids with $N_x = N_y = N_z = 64$ (i.e., $h = 1/64$) and the temporal partition $N \times M = 32 \times 32$ (i.e., $\Delta T = T/N = 5/8, \Delta t = \Delta T/M = 5/256$). We also set the iteration number $k = 32$. The evolutions of the supremum norm and energy of the numerical solutions produced by Parareal-EIFE schemes are presented in Figure 5. We observe that the maximum bound principle is well-preserved for all Parareal-EIFE schemes, and the energy decays monotonically along the time. Figure 6 presents simulated phase structures of the numerical solutions for the Parareal-EIFE22 scheme at times $t = 0, 2.5, 5, 10, 15$, and 20 , from which we can clearly observe the coarsening process of the 3D grains.

6 | CONCLUSIONS

In this article, we develop a parareal method to accelerate the EIFE method for solving semilinear parabolic equations taking the form (1) in regular domains, which is named as the Parareal-EIFE method. The fully discrete solution is obtained by first applying finite element method for spatial discretization and then using explicit exponential Runge–Kutta approach in the parareal framework

for temporal integration. The Parareal-EIFE method not only allows parallel simulation but also fast implementation based on FFT and tensor product for spectral decomposition. For semilinear parabolic equation imposed with the Dirichlet boundary condition, we successfully derive optimal error estimates in the H^1 -norm for the Parareal-EIFE method when its fine and coarse propagators consists of one or two RK stages. Various numerical examples are also presented to illustrate the accuracy and great performance of the proposed Parareal-EIFE method. We also would like to remark that the numerical method and analysis framework developed in this article also naturally enables us to further investigate the localized ETD methods [24, 33] to develop efficient parallel algorithms in both time and space for semilinear parabolic equations.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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